

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

C112529000

I G. Santosh Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

G. Santosh Kumar

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%				

Total Marks 28 /40

Signature of the Faculty [Signature]

Short Answer test

CO1-AT1-Short-answer
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Computer networks

St:-01 file size

$$150 \times 1,048,576 = 157,286,400 \text{ bytes}$$

St:-02 Number of segments

$$\frac{157,286,400}{1500} = 104,857.6$$

St:-03 Number of frames

one segment $\rightarrow$ one frame

$$\Rightarrow 104,858 \text{ frames}$$

St:-04 total overhead

$$104,858 \times 40 = 4,194,320 \text{ bytes}$$

convert to MB

$$\frac{4,194,320}{1,048,576} = 4 \text{ MB}$$

St:-05 total data

$$157,286,400 + 4,194,320 = 161,480,720 \text{ bytes}$$

convert to bits

$$161,480,720 \times 8 = 1,291,845,760 \text{ bits}$$

St: 06

Transmission time

Bandwidth

$$50 \text{ Mbps} = 50,000,000 \text{ bits/sec}$$

Time

$$\frac{1,291,845,760}{50,000,000} = 25.84 \text{ sec}$$

2.

Ans: -

A: 01

no of windows

$$\frac{48}{6} = 8 \text{ windows}$$

St: - 02

Retransmissions

one frame lost in every window

Total windows = 8

Retransmissions = 8

St: 03

useful data sent

$$48 \times 1024 = 49,152 \text{ bytes}$$

IP packet fragmentation

st:01 calculate Maximum Data per fragment
20-byte IP header

$$\text{max Data} = 1500 - 20 = 1480 \text{ bytes.}$$

st:02 calculate no of fragments
original = 12000 bytes

$$\frac{12000}{1480} = 8.108$$

st:03 calculate total header overhead
 $9 \times 20 = 180 \text{ bytes}$

st:04 Total Data Transmitted

$$\text{Original Data} = 12000 \text{ bytes}$$

$$\text{Header overhead} = 180 \text{ bytes}$$

$$\text{Total} = 12000 + 180 = 12180 \text{ bytes}$$

4) sliding window protocol

ex:-

st:-01

number of transmission windows

$$\frac{96}{6} = 8 \text{ transmission windows}$$

st:-02

calculate Retransmission

one frame is lost in every window.

No. of windows = 8

Therefore,

~~8 retransmissions~~

st:-03

Total user Data sent

each frame carries 1024 bytes

$$96 \times 1024 = 99,152 \text{ bytes}$$

$$\text{Total user Data} = 99,152 \text{ bytes}$$